# FAULT EXERCISE

**(1) Formation of the country rock**

**(2) Intrusion of dyke**

Intrusion of a 30 m wide, parallel-sided dyke (discordancy is actually unknown because the orientation of the surfaces in the country rock are unknown), contacts 165/55E, with an unknown opening direction

**(3) Faulting − $F_1$** (crosscuts dyke and hence post-dates it)

Formation of an oblique-slip fault, oriented 030/60SE, slip vector 22/043 and magnitude 230 m (208 m sinistral strike-slip, 97 m normal dip-slip)

**(4) Intrusion of pegmatite** (crosscuts and offsets $F_1$ and hence post-dates it)

Intrusion of a 10 m wide, parallel-sided pegmatite, contacts 110/22N, with an opening direction normal to its margins (in map view), *i.e.*, 020°/200°

**(5) Faulting − $F_2$** (offsets pegmatite and hence post-dates it)

Formation of an oblique-slip fault, oriented 150/60NE, slip vector 25/346 and magnitude 550 m (476 m dextral strike-slip, 275 m reverse dip-slip)

**(6) Uplift and erosion to the present topography**